

Supplementary Material for “Defect-Tolerant Design of Lead-Free RbGeI3 Solar Cells: From SCAPS-1D Optimization to Manufacturing Tolerances, Benchmarked Against CsGeI3”

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September 27, 2026

S1. One-at-a-time sweep details

Full per-parameter traces for the eleven sequential optimization steps summarized in Section V.B of the main text.

Effect of absorber layer thickness

Thickness sets the trade-off between light capture and recombination: a thicker absorber absorbs more photons and lifts J_{SC} , up to the point where the extra volume becomes a recombination liability. PCE peaks at $0.7\ \mu\text{m}$ (20.70%) and eases off beyond it as recombination grows. J_{SC} climbs steadily with thickness, 28.14 to 34.11 mA/cm^2 , while V_{OC} slides from 0.823 to 0.767 V. The $0.7\ \mu\text{m}$ point, where the two effects balance, is the one we keep. The upper end of this J_{SC} range (34.11 mA/cm^2) exceeds the single-gap SQ limit at 1.4 eV ($\approx 33\ \text{mA}/\text{cm}^2$); it was recorded with the early Step-1 deck whose extended sub-gap absorption (cutoff toward $\approx 1.2\ \text{eV}$, $SQ \approx 40\ \text{mA}/\text{cm}^2$) accommodates it. Re-simulation of the thickness sweep with the consolidated uniform-1.4 eV deck gives J_{SC} 25.2–31.3 mA/cm^2 (all below SQ) with the same peak-plateau shape (plateau 0.7–1.0 μm within 0.06 points), so the $0.7\ \mu\text{m}$ choice stands.

Effect of absorber dielectric constant

The dielectric constant barely matters. Raising ϵ_r from 15 to 23.1 moves PCE from 19.92% to 19.79%, a shift carried almost entirely by V_{OC} (0.819 to 0.812 V); J_{SC} and FF are essentially unchanged. With nothing to gain at the high end, we keep $\epsilon_r = 15$.

Effect of absorber electron affinity

Electron affinity lines up the energy bands between adjacent layers and so controls extraction of the charge carriers. We held everything else fixed and stepped χ from 3.9 to 4.2 eV. Peak performance lands at $\chi = 3.9\ \text{eV}$ with a PCE of 19.79%. Raise the affinity to 4.2 eV and V_{OC} falls from 0.812 to 0.710 V while J_{SC} holds flat: the extra electron affinity buys a fill-factor gain smaller than the voltage it costs. We stay at 3.9 eV.

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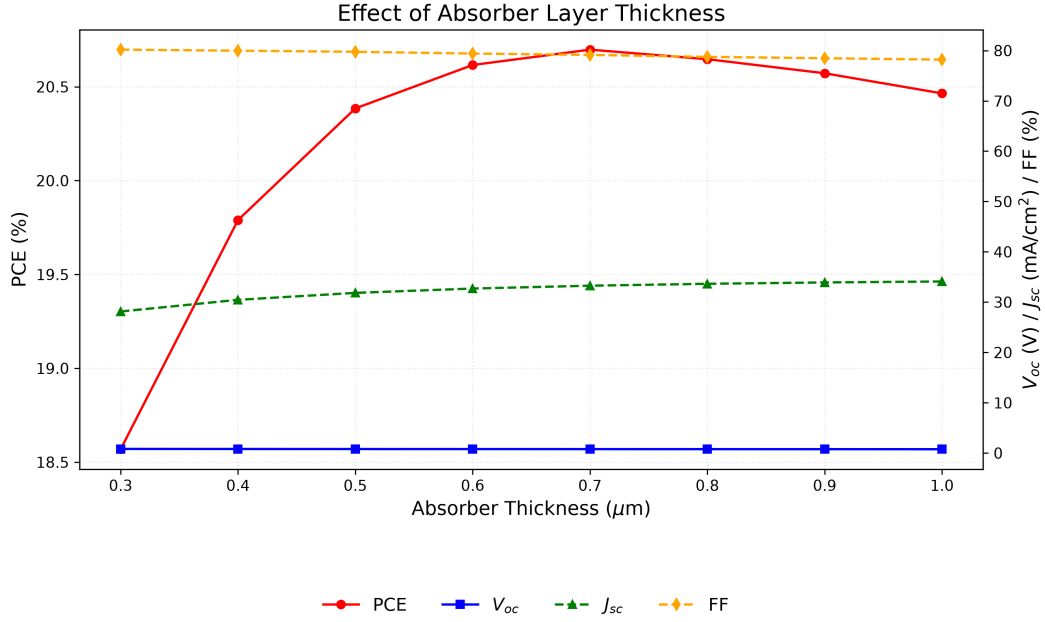


Figure S1: Variation of PCE, V_{OC} , J_{SC} , and FF with RbGeI₃ absorber thickness. PCE peaks at 0.7 μm , beyond which recombination losses reduce efficiency.

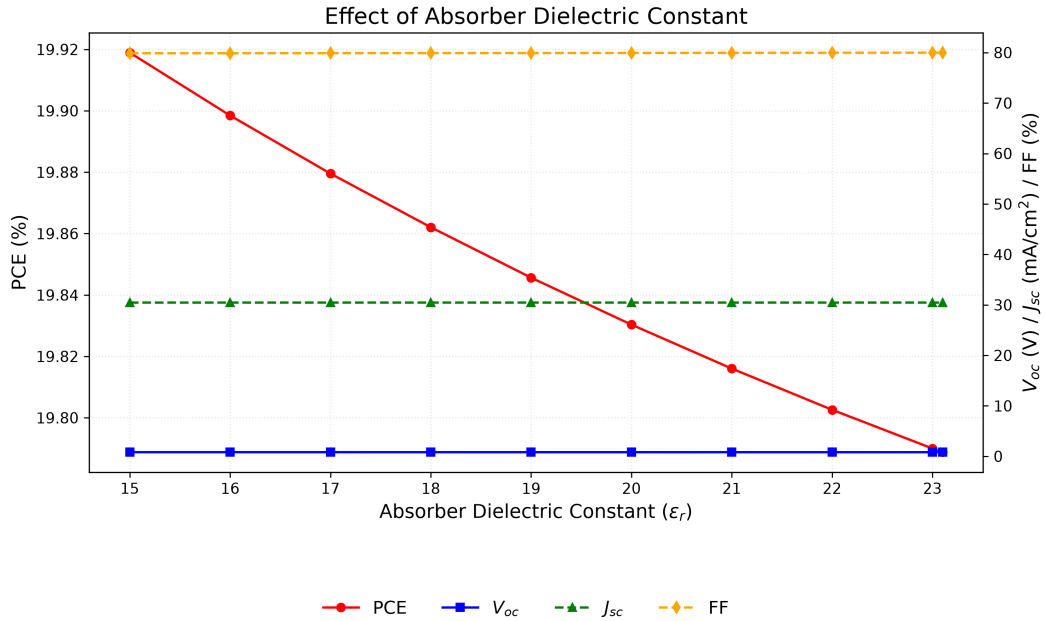


Figure S2: Variation of PCE, V_{OC} , J_{SC} , and FF with RbGeI₃ absorber dielectric constant. The performance is essentially insensitive to ϵ_r across the investigated range; $\epsilon_r = 15$ is used in the final device.

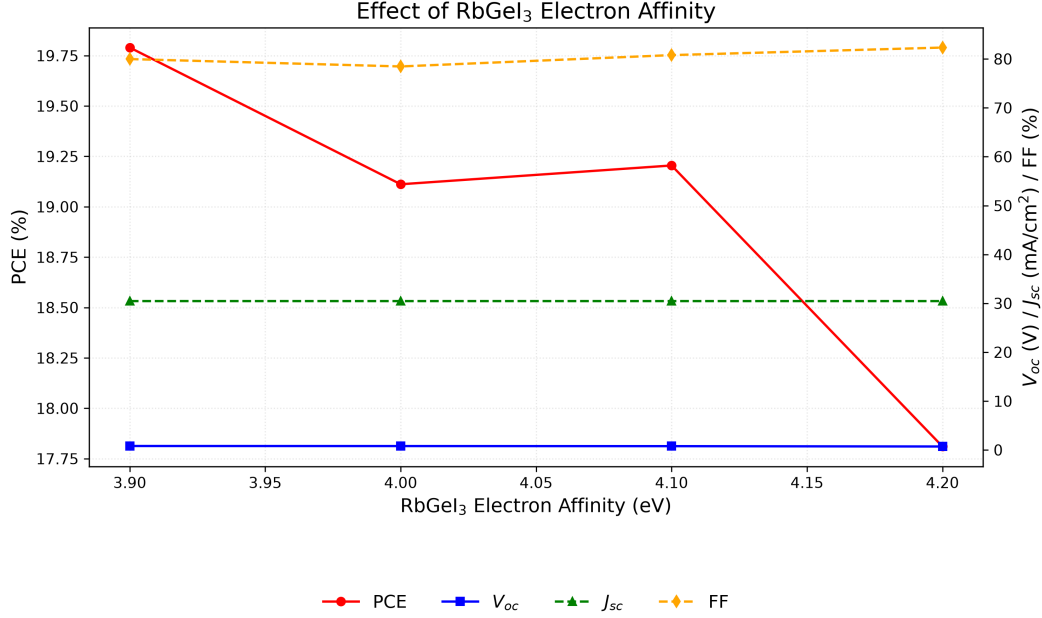


Figure S3: Variation of PCE, V_{OC} , J_{SC} , and FF with RbGeI₃ absorber electron affinity. PCE is highest at $\chi = 3.9$ eV due to optimal band alignment with TiO₂.

Effect of TiO₂/RbGeI₃ interfacial defect density

The TiO₂/RbGeI₃ interface is where performance is won or lost. Varying its defect density from 10^{12} to 10^{18} cm⁻², with everything else fixed, sends the device off a cliff: PCE drops from 21.04% to 8.40%, V_{OC} from 0.827 to 0.439 V, J_{SC} from 30.46 to 28.09 mA/cm², and FF from 83.52% to 68.11% (Figure S4). More states at the TiO₂/RbGeI₃ junction mean more carriers captured before they reach the electrodes; V_{OC} , J_{SC} , and FF fall together. Best performance comes at the unphysical 10^{12} cm⁻² floor, but the final device targets 10^{14} cm⁻², a density a real film can hit and keep: Ge²⁺ oxidation and moisture sensitivity during preparation make anything below that hard to hold consistently. The figure also matches the bulk density chosen in Section V.C and the value comparable SCAPS studies of lead-free perovskites assume.

Effect of RbGeI₃/CuI interfacial defect density

The RbGeI₃/CuI junction is far more tolerant: PCE moves only from 19.79% to 18.19% across the same 10^{12} to 10^{18} cm⁻² range. That contrast pins the performance limit on the TiO₂/RbGeI₃ interface. The final device still adopts 10^{14} cm⁻² here to match the electron side (Supplementary Section S1.4); the modest penalty is the price of internal consistency.

Effect of absorber bandgap

The bandgap picks which wavelengths the cell can use. Moving it from 1.3 to 1.7 eV, all else constant, raises V_{OC} from 0.840 V to 1.002 V as expected from larger quasi-Fermi-level splitting, while J_{SC} falls from 33.59 to 20.30 mA/cm² because a wider gap throws away more of the spectrum. The maximum thus sits at 1.4 eV, within 0.06 eV of the Shockley–Queisser (SQ) optimum of ≈ 1.34 eV for a single junction [1, 2, 3], so the absorber runs at 1.4 eV; tabulated SQ limits imply an ideal

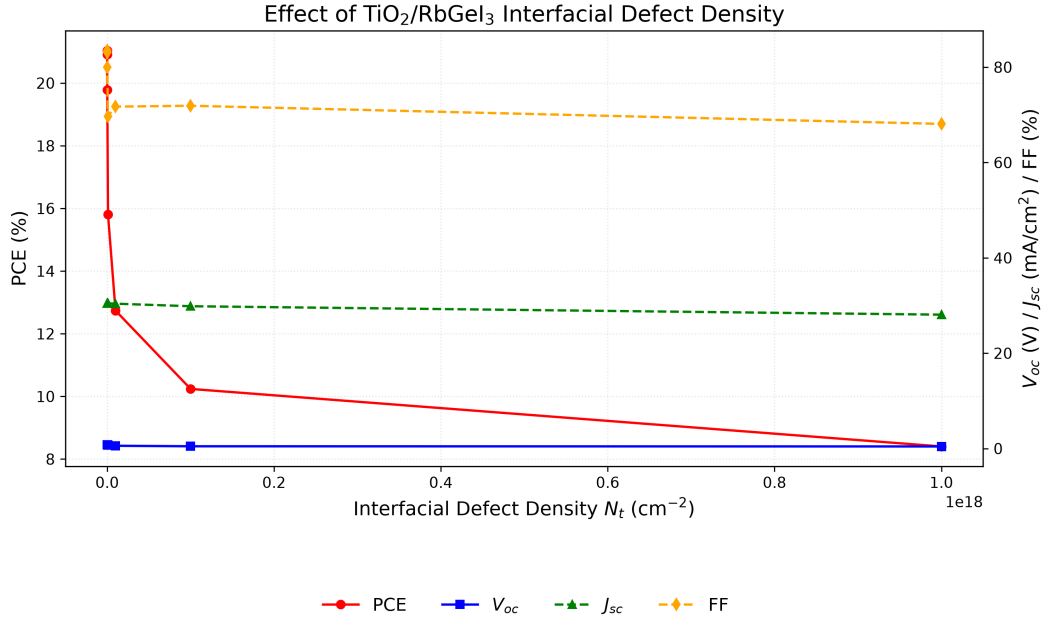


Figure S4: Variation of PCE, V_{OC} , J_{SC} , and FF with TiO₂/RbGeI₃ interfacial defect density. Performance degrades sharply above 10^{14} cm⁻²; the RbGeI₃/CuI interface (Supplementary Section S1.5) is markedly more defect-tolerant.

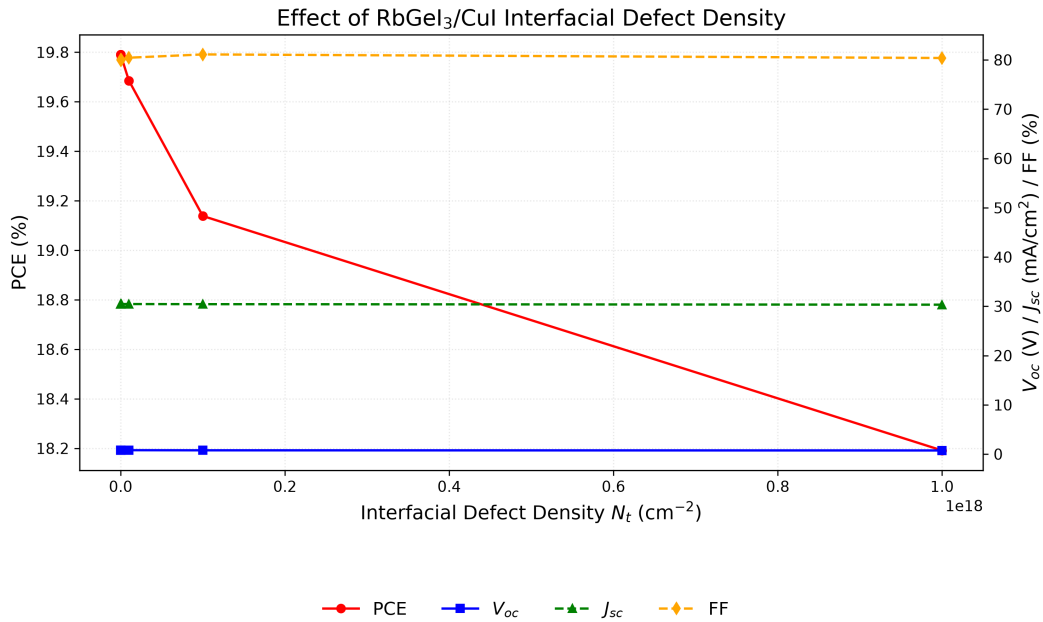


Figure S5: Variation of PCE, V_{OC} , J_{SC} , and FF with RbGeI₃/CuI interfacial defect density. The interface is markedly more defect-tolerant than the TiO₂/RbGeI₃ interface; 10^{14} cm⁻² is used here too.

J_{SC} of ≈ 33 mA/cm² and a maximum efficiency of $\approx 33\%$ at this bandgap [2]. On both sides of the SQ optimum the limiting efficiency declines, steeply for wider bandgaps because fewer photons are absorbed [2, 3]: a 2.0 eV absorber would shift its absorption edge to $\lambda_c \approx 620$ nm and exclude most of the visible spectrum, and our sweep already shows J_{SC} falling to 20.30 mA/cm² at 1.7 eV, whereas at 1.3 eV the higher J_{SC} (33.59 mA/cm²) does not offset the V_{OC} loss (0.840 V). The corresponding cutoff wavelength is $\lambda_c = 1240 / 1.4 \approx 886$ nm, consistent with the absorption edge observed in the quantum-efficiency spectrum of Section V.D of the main text. A targeted re-simulation varying only the absorption-grading endpoints (graded 885.7–1033.3 nm versus flat 885.7–885.7 nm, all else fixed) changes J_{SC} by 0.00 mA/cm², and the consolidated J_{SC} of 30.32 mA/cm² lies below the SQ limit of ≈ 33 mA/cm² at 1.4 eV, so the headline result is admissible with uniform square-root-law absorption at the electrical bandgap; the 26.81% device is therefore reported as an ideal-coupling upper bound, with a quantified realistic window given in Sections V.G and VII.A of the main text.

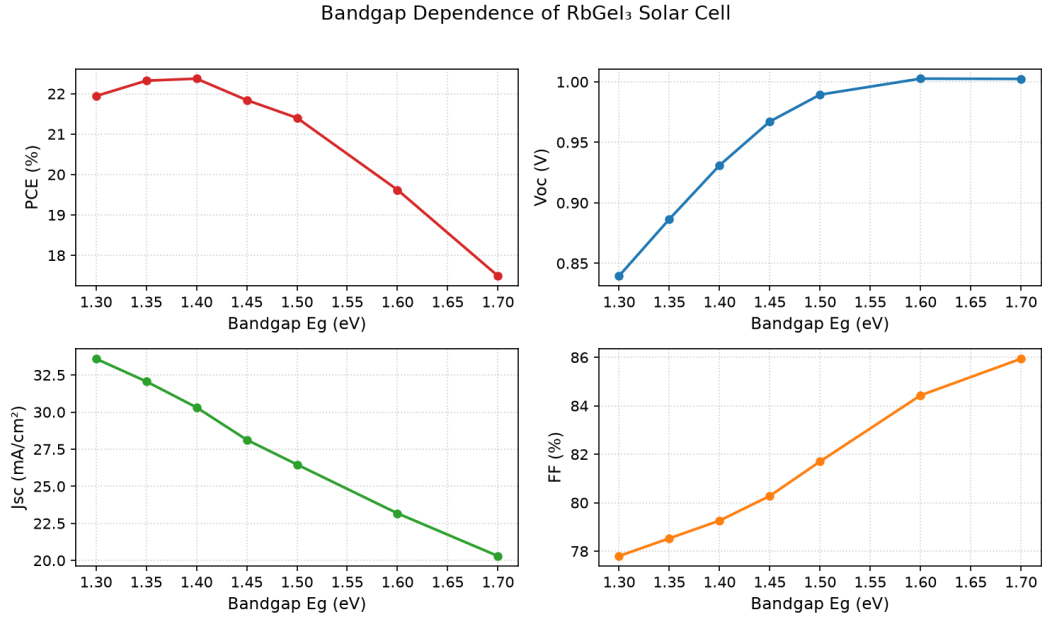


Figure S6: Variation of PCE, V_{OC} , J_{SC} , and FF with RbGeI₃ absorber bandgap. PCE peaks at 1.4 eV, close to the Shockley–Queisser optimum.

Effect of TiO₂ ETL thickness

Thickness of the TiO₂ ETL cuts across electron extraction, transport, and optical transmission at once. Sweeping it from 0.01 to 0.09 μm , all else fixed, gives the best PCE at the thinnest layers: 21.12% at 10 nm, a local low of 18.72% at 0.04 μm , and a partial recovery to 19.97% at 0.09 μm . Planar SCAPS studies stick to thin (≤ 50 nm) ETLs for the same reason we do: thicker ones add series resistance and soak up photons.

Effect of CuI HTL thickness

CuI thickness, swept from 0.1 to 1.9 μm , moves PCE by less than 0.002 percentage points; V_{OC} , J_{SC} , and FF stay essentially constant. That flatness lets us pick 100 nm purely to minimize material

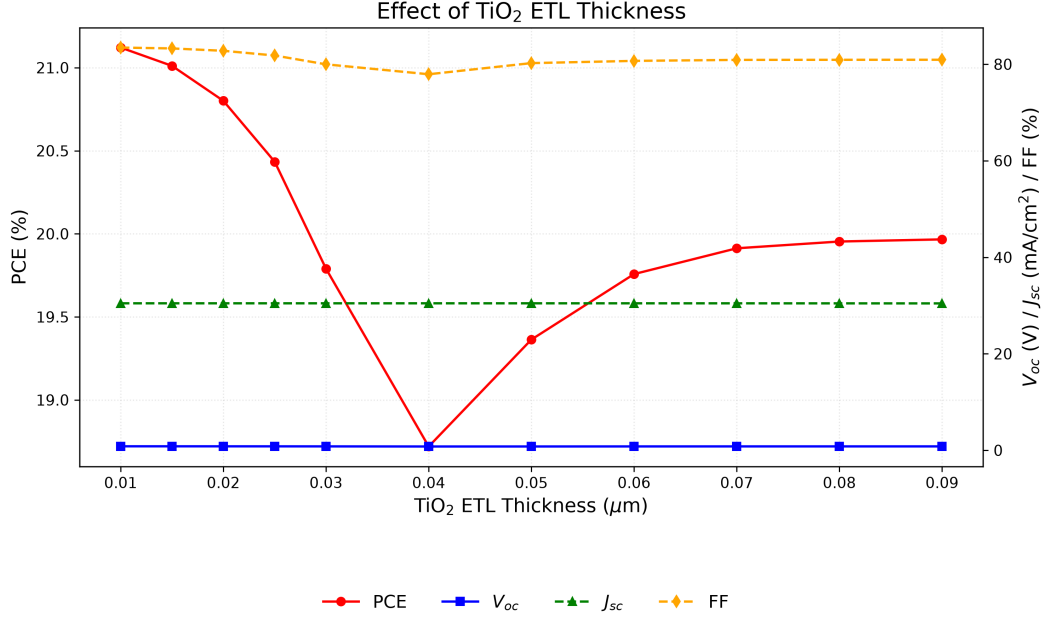


Figure S7: Variation of PCE, V_{OC} , J_{SC} , and FF with TiO_2 ETL thickness. PCE is highest at the thinnest ETL (10 nm), reflecting the shorter electron transport path.

use. It sits at the low end of the 0.1–1.0 μm range comparable studies explored, which also reported a weak dependence.

Effect of conduction-band effective density of states N_C

Raising N_C from 1×10^{17} to $1 \times 10^{20} \text{ cm}^{-3}$ pushes PCE monotonically down from 20.65% to 18.15%; V_{OC} carries the loss (0.859 to 0.764 V) while J_{SC} barely stirs at $\sim 30.46 \text{ mA/cm}^2$. A lower N_C means a lower intrinsic carrier density and therefore a higher V_{OC} , so the bottom of the sweep, $1 \times 10^{17} \text{ cm}^{-3}$, goes into the final device. SCAPS studies commonly assume $\approx 10^{18} \text{ cm}^{-3}$ for lead-free absorbers; we find going lower helps.

Effect of valence-band effective density of states N_V

N_V behaves like N_C but hits harder. Across 1×10^{17} to $1 \times 10^{20} \text{ cm}^{-3}$, PCE drops from 23.54% to 18.86% as V_{OC} slides from 0.945 to 0.780 V with J_{SC} flat. The smallest value wins again: $N_V = 1 \times 10^{17} \text{ cm}^{-3}$ is carried into the final set.

S2. Illumination and dark characteristics

Illumination dependence

The optimized cell was also driven under 0.1 to 1.5 suns (Figure S11). J_{SC} tracks intensity almost linearly, the natural consequence of carrier generation scaling with photon flux, while V_{OC} rises logarithmically. The slope of that logarithmic rise gives an ideality factor $n \approx 1.17$, so deep-defect Shockley-Read-Hall recombination is not dominant under operating conditions, and the cell stays efficient across the entire range. Re-simulation (0.1–1.5 suns, archived illum_results.json) gives J_{SC}

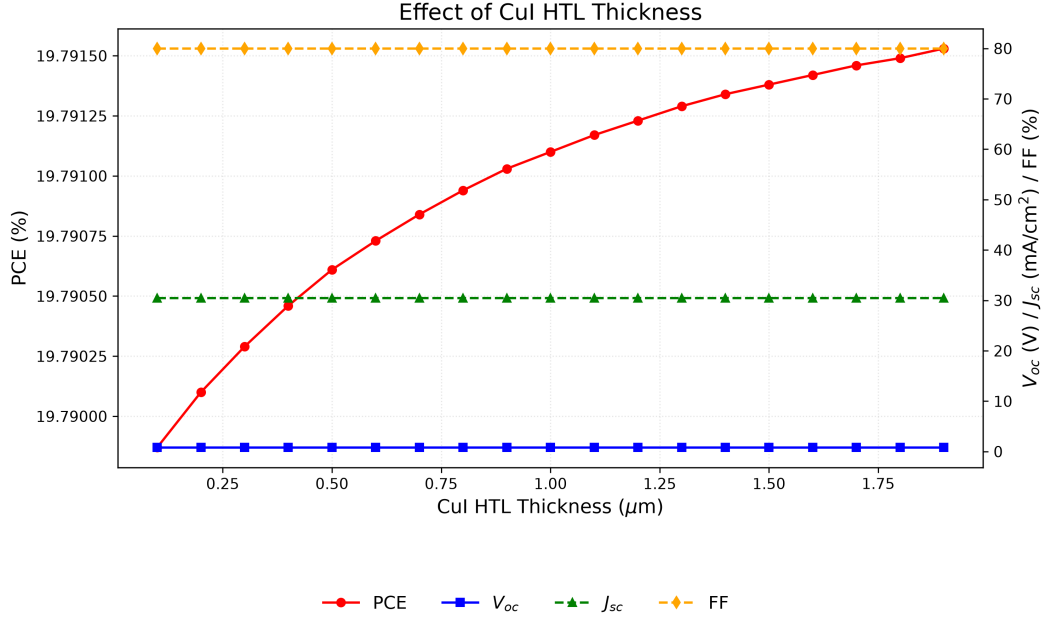


Figure S8: Variation of PCE, V_{OC} , J_{SC} , and FF with CuI HTL thickness. The performance is essentially independent of HTL thickness; 100 nm minimizes material usage.

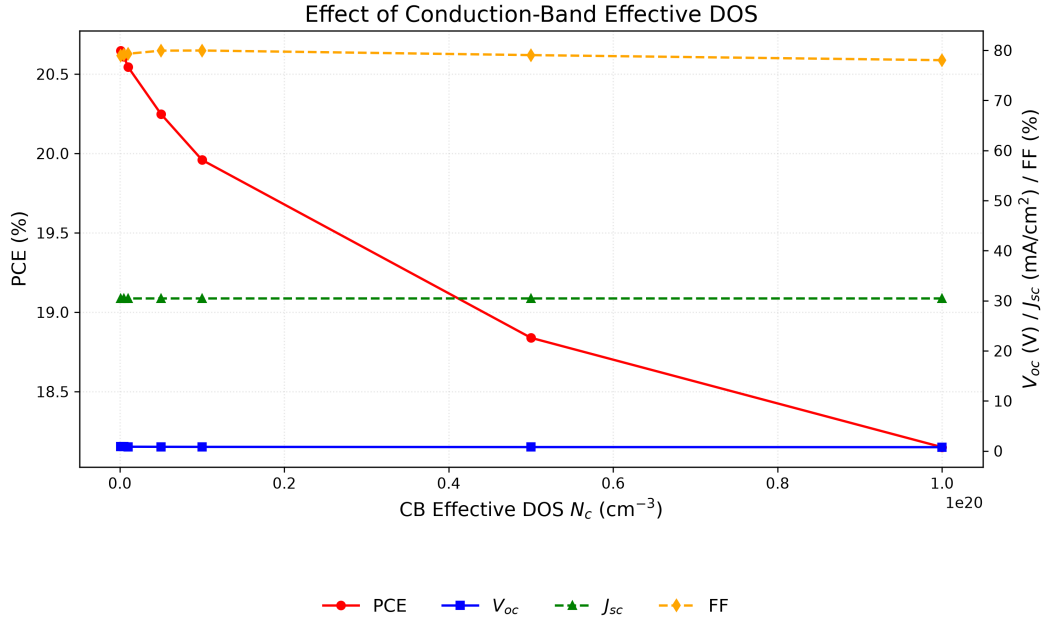


Figure S9: Variation of PCE, V_{OC} , J_{SC} , and FF with conduction-band effective density of states N_C . Lower N_C is preferred; the best value is 10^{17} cm^{-3} .

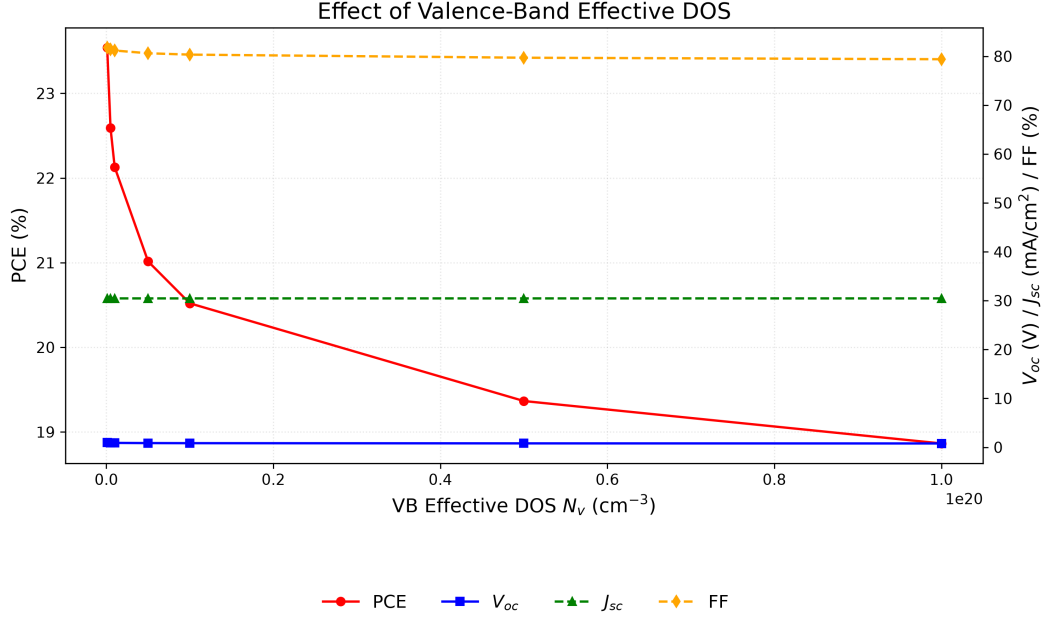


Figure S10: Variation of PCE, V_{OC} , J_{SC} , and FF with valence-band effective density of states N_V . The effect is more pronounced than for N_C ; the best value is 10^{17} cm^{-3} .

= $30.32 \times \text{suns} + 0.00$ (perfect linearity) and $n = 1.17$ from the V_{OC} - $\ln(I)$ slope, confirming the reported $n \approx 1.2$.

Dark J–V characteristics

Under dark bias (Figure S12) the cell rectifies hard: forward current at +1 V exceeds reverse current at -0.5 V by a factor of 1.8×10^{10} , and leakage stays under $10^{-10} \text{ A}/\text{cm}^2$ across the whole reverse window (-0.5 to $+1.3 \text{ V}$). The exponential forward region fits to an ideality factor of $n \approx 1.5$ and a reverse saturation current $J_0 \approx 5 \times 10^{-11} \text{ A}/\text{cm}^2$. Together these numbers describe high-quality bulk and interfaces with no measurable shunts.

References

- [1] W. Shockley and H. J. Queisser, “Detailed balance limit of efficiency of p-n junction solar cells,” J. Appl. Phys. 32(3), 510–519 (1961). doi:10.1063/1.1736034.
- [2] S. Rühle, “Tabulated values of the Shockley–Queisser limit for single junction solar cells,” Sol. Energy 130, 139–147 (2016). doi:10.1016/j.solener.2016.02.015.
- [3] C. H. Henry, “Limiting efficiencies of ideal single and multiple energy gap terrestrial solar cells,” J. Appl. Phys. 51(8), 4494–4500 (1980). doi:10.1063/1.328272.

S3. Screening and simulation setup tables

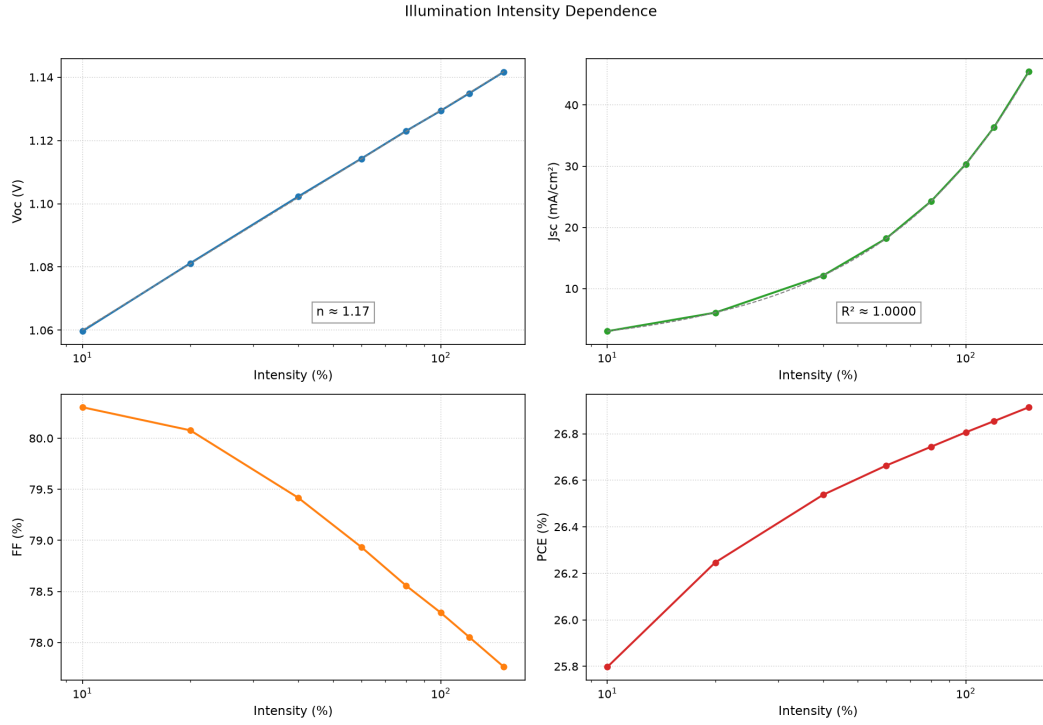


Figure S11: Variation of photovoltaic parameters of the optimized device with illumination intensity from 0.1 to 1.5 suns.

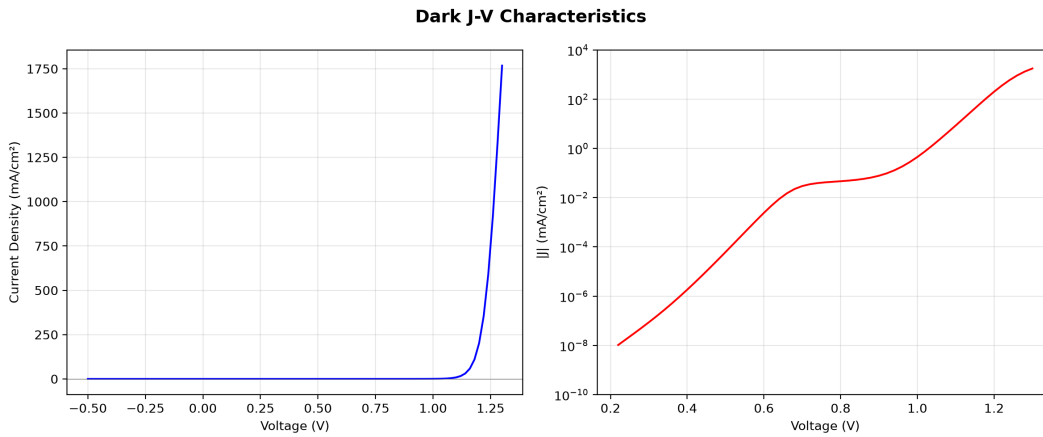


Figure S12: Dark J-V characteristic of the optimized device. The forward branch is exponential over nearly three decades; the reverse branch remains below 10^{-10} A/ cm^2 .

Study (Year) [Ref.]	Device Structure	PCE (%)	Key Limitation
Pindolia et al. (2022) [?]	FTO/TiO ₂ /RbGeI ₃ /NiO/Ag	9.77	Broad multi-parameter sweep (HTL, ETL, thickness, doping, defects, back contact, temperature); baseline efficiency remains modest and Ge ²⁺ oxidation not addressed.
Saikia et al. (2022) [?]	FTO/TiO ₂ /CsGeI ₃ /CuI/Au	10.80	No multi-HTL comparative analysis.
Bouazizi et al. (2022) [?]	FTO/PCBM/ CsSn _{0.5} Ge _{0.5} I ₃ /Spiro/Au	18.13	Mixed Ge/Sn absorber; thermal sensitivity.
Loumachi et al. (2024) [?]	ITO/C ₆₀ /RbGeI ₃ /CBTS/Ag	27.21	Optimized RbGeI ₃ thickness and series/shunt resistance; interface defect optimization not addressed.
Raj et al. (2025) [?]	FTO/ TiO ₂ /RbGeI ₃ /PEDOT:PSS/ Au	18.60	Organic HTL limits long-term stability.
Pathak et al. (2026) [?]	FTO/TiO ₂ /RbGeI ₃ /Spiro- OMeTAD/Au	26.47	Ge ²⁺ → Ge ⁴⁺ oxidation not addressed; only single ETL/HTL combination studied.
Dash et al. (2025) [?]	DFT + SCAPS-1D of RbGeX ₃ (X=Cl, Br, I)	25.76	Primarily material-level comparison; limited device-architecture optimization.

Table 1: Table S1. Summary of recent SCAPS-1D studies on lead-free Ge-based and Sn-based perovskite solar cells.

Parameter	TiO ₂	FTO	RbGeI ₃	CuI
Thickness (nm)	30	500	400	100
Bandgap E_g (eV)	3.2	3.2	1.4	3.1
Electron affinity χ (eV)	4.0	4.4	3.9	2.1
Dielectric permittivity ϵ_r	9	9	23.01	6.5
CB effective DOS N_C (cm ⁻³)	2×10^{18}	2.2×10^{18}	1.4×10^{19}	2.8×10^{19}
VB effective DOS N_V (cm ⁻³)	1.8×10^{19}	1.8×10^{19}	2.8×10^{19}	1×10^{19}
Electron mobility μ_e (cm ² V ⁻¹ s ⁻¹)	20	20	28.6	100
Hole mobility μ_p (cm ² V ⁻¹ s ⁻¹)	10	10	27.3	43.9
Donor density N_D (cm ⁻³)	9×10^{16}	1×10^{19}	1×10^9	0
Acceptor density N_A (cm ⁻³)	0	0	1×10^9	1×10^{18}
Bulk defect density N_t (cm ⁻³)	1×10^{15}	1×10^{14}	1×10^{15}	1×10^{15}

Table 2: Table S2. Initial electrical and optical parameters used in the SCAPS-1D simulation of the proposed FTO/TiO₂/RbGeI₃/CuI/Au solar cell.

Interface	Interface defect density N_t (cm ⁻²)
TiO ₂ / RbGeI ₃	1×10^{14}
RbGeI ₃ / CuI	1×10^{14}

Table 3: Table S3. Initial interface defect density applied at each heterojunction.

Parameter	Value
Simulation software	SCAPS-1D version 3.3.10
Illumination spectrum	AM 1.5G
Incident power density	100 mW/cm ² (1000 W/m ²)
Operating temperature	300 K (27 °C)
Illumination intensity	1 Sun
Scan mode	Illuminated J–V
Device type	Planar heterojunction
Front contact	FTO
Back contact	Au
Initial material parameters	Table II
Initial interface defect density	Table III

Table 4: Table S4. Simulation conditions used in SCAPS-1D.